

Home Service Booking Platform

Microservices API Documentation

This document lists all REST APIs used in the Home Service Booking Platform. The system consists of 3 independent microservices, each with its own database and dedicated port.

Architecture Overview

Technology Stack:

- Backend: Java Spring Boot 4.1.0
- Database: MySQL / MariaDB (XAMPP)
- Architecture: 3 Microservices (Database per Service)
- Inter-service Communication: REST API (WebClient)

Service Summary

#	Microservice	Port	Database	Total APIs
1	Provider	8082	homeservice_provider_db	10 APIs
2	Booking	8081	homeservice_booking_db	11 APIs
3	Notification	8083	homeservice_notification_d	6 APIs

1. Provider Microservice

Base URL: http://localhost:8082

Database: homeservice_provider_db

Purpose: Manages service providers and service categories.

#	Method	Endpoint	Description
1	GET	/api/categories	Get all service categories
2	GET	/api/categories/{id}	Get category by ID
3	GET	/api/providers	Get all providers
4	GET	/api/providers/available	Get available providers only
5	GET	/api/providers/{id}	Get provider by ID
6	GET	/api/providers/category/{categoryId}	Get providers by category
7	GET	/api/providers/{id}/exists	Check if provider exists
8	POST	/api/providers	Register new provider
9	POST	/api/providers/login	Provider login
10	PUT	/api/providers/{id}/availability	Update provider availability

Sample Request - Register Provider (POST /api/providers)

```
{
  "categoryId": 1,
  "firstName": "John",
  "lastName": "Silva",
  "email": "john@gmail.com",
  "password": "123456",
  "phone": "0771234567",
  "location": "Colombo",
  "experienceYears": 5
}
```

2. Booking Microservice

Base URL: `http://localhost:8081`

Database: `homeservice_booking_db`

Purpose: Manages customers, bookings, and payments.

#	Method	Endpoint	Description
1	POST	<code>/api/customers/register</code>	Register new customer
2	POST	<code>/api/customers/login</code>	Customer login
3	GET	<code>/api/customers/{id}</code>	Get customer by ID
4	POST	<code>/api/bookings</code>	Create new booking
5	GET	<code>/api/bookings/{id}</code>	Get booking by ID
6	GET	<code>/api/bookings/customer/{customerId}</code>	Get bookings by customer
7	GET	<code>/api/bookings/provider/{providerId}</code>	Get bookings by provider
8	PUT	<code>/api/bookings/{id}/status</code>	Update booking status
9	POST	<code>/api/payments</code>	Process payment
10	GET	<code>/api/payments/{id}</code>	Get payment by ID
11	GET	<code>/api/payments/booking/{bookingId}</code>	Get payments by booking

Sample Request - Create Booking (POST `/api/bookings`)

```
{
  "customerId": 1,
  "providerId": 1,
  "bookingDate": "2026-07-20",
  "bookingTime": "10:00:00",
  "serviceAddress": "Colombo",
  "description": "Fix electrical wiring",
  "totalAmount": 5000.00
}
```

Booking Status Values

Pending | Accepted | Rejected | In Progress | Completed | Cancelled

3. Notification Microservice

Base URL: `http://localhost:8083`

Database: `homeservice_notification_db`

Purpose: Sends and manages notifications for customers and providers.

#	Method	Endpoint	Description
1	POST	<code>/api/notifications</code>	Create notification
2	GET	<code>/api/notifications/{id}</code>	Get notification by ID
3	GET	<code>/api/notifications/user/{userId}</code>	Get user notifications (userType param)
4	GET	<code>/api/notifications/user/{userId}/unread</code>	Get unread notifications
5	GET	<code>/api/notifications/booking/{bookingId}</code>	Get notifications by booking
6	PUT	<code>/api/notifications/{id}/read</code>	Mark notification as read

Query Parameter Example

GET `/api/notifications/user/1?userType=Customer`

userType values: Customer | Provider

notificationType values: Booking | Payment | Reminder

Inter-Service Communication

From Service	To Service	When
Booking (8081)	Provider (8082)	Validates provider exists before creating booking
Booking (8081)	Notification (8083)	Sends notification when booking is created
Booking (8081)	Notification (8083)	Sends notification when booking status changes
Booking (8081)	Notification (8083)	Sends notification when payment is processed

How to Run the Backend

1. Start XAMPP MySQL
2. Run stop-services.bat (if ports are busy)
3. Run start-services.bat (opens 3 terminal windows)
4. Wait for 'Tomcat started on port...' in each window
5. Test APIs using Postman

Start Order

Provider (8082) -> Notification (8083) -> Booking (8081)

Test in Postman

1. GET `http://localhost:8082/api/categories`
2. GET `http://localhost:8082/api/providers`
3. POST `http://localhost:8081/api/customers/register`
4. POST `http://localhost:8081/api/bookings`
5. POST `http://localhost:8081/api/payments`
6. GET `http://localhost:8083/api/notifications/user/1?userType=Customer`

HTTP Response Codes

Code	Meaning
200 OK	Successful GET, PUT, login requests
201 Created	Successful POST (register, create, payment)
400 Bad Request	Validation error or invalid input
404 Not Found	Resource not found

Total APIs: 27 | Generated for Home Service Booking Platform